Article

Gauge Structure and Color Confinement in a Deterministic Brane-Lattice Substrate (Paper III)

Lukas Molzberger

Independent researcher; lukas.molzberger@gmx.de

Abstract

We develop the gauge and color consistency bridges for the deterministic 4D brane-lattice substrate of Paper I. The gauge bridge shows that adiabatic transport of a band-isolated wavepacket's polarization eigenbundle induces a Berry connection $a_\mu = \mathrm{i}\langle u|\partial_\mu u\rangle$ that is formally the electromagnetic four-potential A_μ, with Berry curvature $f_{\mu\nu}$ as the field tensor $F_{\mu\nu}$ and the homogeneous Maxwell equations following as the Bianchi identity $\partial_{[\lambda}F_{\mu\nu]} = 0$. The gauge structure does not live in the Brillouin zone: the dynamical matrix is real symmetric and carries identically zero Berry curvature there; we verify this numerically at every α. The physically meaningful gauge connection lives over physical spacetime (x, t), where the $U(1)$ phase is rotation of the carrier envelope along the timelike direction. For a near-degenerate lateral triplet the Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$ generalizes the Berry connection to a non-Abelian gauge potential. The color bridge decomposes the axis-aligned triplet as $\mathfrak{u}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3)$, identifies the $U(1)$ trace sector with electromagnetism and the traceless generators with color, and establishes kinematic confinement: no linear propagation direction simultaneously satisfies coherence (degenerate branches) and color-activity (non-degenerate branches with defined holonomy), so color charge cannot be carried to infinity by a free linear excitation. The falsifiable prediction is the off-[111] fibre-holonomy ratio $R(\alpha_2)/R(\alpha_1) = [(3 - 2\alpha_2)/\alpha_2]/[(3 - 2\alpha_1)/\alpha_1]$, equal to 0.3077 for $(\alpha_1, \alpha_2) = (0.2, 0.5)$.

Keywords: Berry connection; Wilczek–Zee connection; gauge emergence; color confinement; kinematic confinement; U(3); SU(3); brane lattice; band isolation; semilocal vortex; Skyrmion

1. Introduction

This is the third paper in a series developing a deterministic brane-lattice substrate hypothesis. Paper I established the substrate model and exported the linearized wave-branch structure. Paper II developed the Lorentz and gravity bridges. The present paper develops two closely coupled bridges: gauge structure and color confinement.

The gauge bridge. Each wavepacket carries a local polarization state, defined as a normalized eigenvector of the linearized carrier operator about the slowly varying background at the wavepacket's location. As the wavepacket is transported through regions of varying background, the eigenvector is parallel-transported along the connection it defines. This is the Berry connection (Section 3): for a single isolated mode, $a_\mu = \mathrm{i}\langle u|\partial_\mu u\rangle$ is formally the electromagnetic four-potential A_μ, its curvature $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ is the field tensor $F_{\mu\nu}$, and the homogeneous Maxwell equations follow as the Bianchi identity $F = \mathrm{d}A$. No new

Received:
Accepted:
Published:

https://doi.org/10.3390/sym1010000

assumption is added: gauge structure is not a coincidence but the canonical outcome of transporting a polarization eigenbundle.

A key point developed in Section 3.5: the gauge connection does *not* live in the Brillouin zone. The dynamical matrix $D(\mathbf{k})$ is real symmetric and carries identically zero Berry curvature over the BZ at every α; we have verified this numerically. The physically relevant gauge connection lives over physical spacetime (x, t), where the $U(1)$ phase is the rotation of the carrier field as one advances along the timelike direction.

The color bridge. The cubic substrate provides a natural three-dimensional polarization sector: the axis-aligned lateral triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$. For a near-degenerate triplet (small α), adiabatic transport induces a Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$. Decomposing $\mathfrak{u}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3)$ identifies the $U(1)$ trace sector with electromagnetism and the eight traceless generators with a color-like internal degree of freedom. The color bridge (Section 4) then establishes a structural fact: *kinematic confinement*. No free linear propagation direction simultaneously satisfies coherence (degenerate branches) and color-activity (non-degenerate branches with defined internal holonomy), so color charge cannot be carried to infinity by any free linear excitation. Color is structurally confined to nonlinear bound states — without invoking a separate QCD action.

Why gauge and color are paired. The two bridges share a common foundation: per-wavepacket band isolation of the lateral triplet. The gauge bridge extracts the Abelian rank-1 sector; the color bridge extracts the non-Abelian triplet sector. Both structures emerge from the same eigenbundle transport, and the $U(3)$ decomposition that distinguishes them is fixed by the cubic lattice geometry, not introduced by hand. Pairing them in one paper makes this shared derivation explicit.

Paper organization. Section 2 lists the interface equations imported from Paper I. Section 3 develops the gauge bridge: per-wavepacket isolation, Berry and Wilczek–Zee connections, the EM identification, where the gauge structure lives, and the spinorial holonomy. Section 4 develops the color bridge: axis-triplet degeneracy, $U(3)$ decomposition, topological sectors, kinematic confinement, and the numerical targets.

2. Imported interfaces from Paper I

This paper imports the following results from Paper I without re-derivation. All equation and section labels without an explicit paper prefix refer to Paper I.

Substrate model and equations of motion. The 4D brane embedding, induced metric g_{ij} (Equation (10)), constitutive law (Section 3.5), and equations of motion (Equation (26)).

Linearized branch structure. The long-wavelength dispersion $\omega_b^2(\mathbf{k}) \approx c_b^2 |\mathbf{k}|^2$ (Equation (32)), branch speeds $c_T^2 = (1-\alpha) k_s a^2 / m$ and $c_L^2 = k_s a^2 / m$ (Equation (17)).

Axis-aligned triplet structure. The three spacelike embedding components (X^1, X^2, X^3) on the 6-neighbor cubic lattice define an axis-aligned triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$, with α-driven degeneracy described in Section 4 of Paper I.

Geometric nonlinearity. The constitutive quartic coefficient $\propto k_s \alpha / a$ from Equation (20), which sets the energetic coupling between the $U(1)$ and $SU(3)$ sectors at finite amplitude.

Bell interpretive stance. Retrocausal worldtube reading from Section 5 of Paper I, inherited as a fixed commitment.

https://doi.org/10.3390/sym1010000

3. Gauge bridge: Berry/Wilczek–Zee connections as gauge structure

3.1. Per-wavepacket band isolation and polarization transport

The complex-envelope description is applied per band-isolated excitation, not to a globally coherent carrier sector spanning macroscopic scales. Each wavepacket is treated as narrowband around its own local carrier angular frequency ω_0, determined by the linearized carrier operator about the slowly varying background at the wavepacket's location. A band-isolated wavepacket can be written locally as

$$\mathbf{u}(x,t) \;\approx\; \mathrm{Re}\Big\{a(x,t)\,\boldsymbol{\epsilon}(x,t)\,e^{\mathrm{i}\theta(x,t)}\Big\}, \qquad \theta(x,t) = \phi(x,t) - \omega_0 t, \tag{1}$$

where $a(x,t) \geq 0$ is the slowly varying envelope amplitude, $\boldsymbol{\epsilon}(x,t) \in \mathbb{C}^N$ is a normalized polarization state ($\boldsymbol{\epsilon}^\dagger\boldsymbol{\epsilon} = 1$), and ω_0 is the wavepacket's local carrier angular frequency.

A central point is that polarization transport is not unique: the same physical state can be represented in many local phase conventions. Geometric phase formalizes this redundancy as a connection on a bundle of polarization states over the spacetime manifold parameterizing the wavepacket's trajectory.

3.2. Rank-1 Berry connection and electromagnetic identification

For a single isolated polarization mode, represent the local state by a normalized vector $|u(x)\rangle \in \mathbb{C}^N$ that depends smoothly on spacetime position x^μ (with $x^\mu = (ct, x^1, x^2, x^3)$ built from the intrinsic coordinates and a chosen branch speed c). Define the Berry connection

$$a_\mu(x) \;:=\; \mathrm{i}\,\langle u(x)\,|\,\partial_\mu u(x)\rangle. \tag{2}$$

Under a local phase choice $|u(x)\rangle \mapsto e^{-\mathrm{i}\chi(x)}|u(x)\rangle$,

$$a_\mu \;\mapsto\; a_\mu + \partial_\mu\chi, \tag{3}$$

so only gauge-invariant quantities carry physical meaning. The Berry curvature is

$$f_{\mu\nu} \;:=\; \partial_\mu a_\nu - \partial_\nu a_\mu, \tag{4}$$

and the holonomy around a closed loop Γ is

$$\gamma_\Gamma = \oint_\Gamma a_\mu\,\mathrm{d}x^\mu = \int_{\Sigma:\,\partial\Sigma=\Gamma} f_{\mu\nu}\,\mathrm{d}\Sigma^{\mu\nu}. \tag{5}$$

Electromagnetic reading: four-potential and field tensor. In the $U(1)$ trace sector (Section 4.2) the Berry connection (2) *is* the electromagnetic four-potential, $A_\mu \equiv a_\mu$, and its curvature (4) *is* the field tensor $F_{\mu\nu} \equiv f_{\mu\nu}$, with $E_i = F_{0i}$ (time–space) and $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$ (space–space). Two consequences are automatic, requiring no field dynamics:

1. the gauge freedom (3) is exactly the unobservable choice of carrier phase reference, and
2. the homogeneous Maxwell equations are the Bianchi identity $\partial_{[\lambda}F_{\mu\nu]} = 0$, which holds identically because $F = \mathrm{d}A$.

A nonzero field ($F \neq 0$) requires the phase to be non-integrable: the polarization state $|u\rangle$ must genuinely rotate within $\mathbb{C}^N$ ($N \geq 2$), not merely carry an overall phase, since a pure gradient $A_\mu = \partial_\mu\chi$ is pure gauge. What this kinematics does *not* fix is field *dynamics*: the inhomogeneous equations $\mathrm{d}{\star}F = J$ hold only if the substrate furnishes a Maxwell action $-\frac{1}{4}\int F \wedge {\star}F$, a separate open derivation.

3.3. Eigenbundle picture and adiabaticity requirement

The connection (2) is not attached to an arbitrary amplitude/phase split. The physical choice is $|u(x)\rangle$ as a locally defined eigenvector of the linearized carrier operator about the slowly varying background at x. As x varies, these eigenvectors form an eigenbundle, and the Berry/Wilczek–Zee connection is the natural connection on that bundle — the same eigenbundle picture that underlies Berry-phase effects in crystalline electronic systems [1], here applied to the polarization eigenbundle of the mechanical carrier.

A clean rank-1 ($U(1)$) reduction requires dynamics to remain in a single isolated band. If $\Delta\omega_{\text{gap}}(x)$ is the local gap to the nearest competing band and $\Omega_{\text{mix}}(x)$ is the non-adiabatic mode-conversion rate induced by background gradients, the adiabatic condition

$$\Omega_{\text{mix}}(x) \ll \Delta\omega_{\text{gap}}(x) \tag{6}$$

must hold on the envelope scale. When (6) fails, the correct reduction retains the relevant multi-dimensional subspace (the Wilczek–Zee description). The prestress parameter α and wavelength regime jointly control branch separation and therefore the validity of the Abelian gauge description.

3.4. Wilczek–Zee connection for the degenerate triplet

For an n-dimensional near-degenerate subspace, choose an orthonormal frame $\mathbf{E}(x) \in \mathbb{C}^{N\times n}$ with columns $|u_a(x)\rangle$ spanning the subspace ($\mathbf{E}^\dagger\mathbf{E} = I_n$). The Wilczek–Zee connection is

$$\mathcal{A}_\mu(x) \;:=\; \mathrm{i}\,\mathbf{E}(x)^\dagger\,\partial_\mu\mathbf{E}(x) \;\in\; \mathfrak{u}(n). \tag{7}$$

Under $\mathbf{E} \mapsto \mathbf{E}U(x)$ with $U(x) \in U(n)$,

$$\mathcal{A}_\mu \;\mapsto\; U^\dagger\mathcal{A}_\mu U + \mathrm{i}\,U^\dagger\partial_\mu U, \tag{8}$$

and the non-Abelian curvature is

$$\mathcal{F}_{\mu\nu} \;:=\; \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu - \mathrm{i}[\mathcal{A}_\mu, \mathcal{A}_\nu]. \tag{9}$$

For $n = 3$ (the lateral triplet, Section 4), $\mathcal{A}_\mu \in \mathfrak{u}(3)$ decomposes into a $U(1)$ trace part and a traceless $SU(3)$ part, with the commutator term in (9) carrying the self-coupling of the non-Abelian sector.

Gauge-covariant envelope dynamics. Projecting the full substrate equations of motion onto the retained n-dimensional subspace produces a gauge-covariant derivative acting on the slow envelope $\Psi \in \mathbb{C}^n$:

$$\mathcal{D}_\mu\Psi \;=\; (\partial_\mu + \mathrm{i}\,\mathcal{A}_\mu)\Psi, \tag{10}$$

which transforms covariantly under local frame rotations. The coarse-grained slow dynamics then admits a gauge-invariant effective action

$$S_{\text{eff}}[\Psi, \mathcal{A}] = \int \mathrm{d}^4x\,\Big(\Psi^\dagger\mathcal{K}(\mathcal{D})\Psi - \kappa\,\mathrm{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) + \cdots\Big), \tag{11}$$

where $\mathcal{K}(\mathcal{D})$ is determined by the linearized carrier band and κ is an emergent curvature stiffness.

3.5. Where the gauge structure lives: spacetime, not the Brillouin zone

The gauge connection does *not* live in the Brillouin zone. The dynamical matrix $D(\mathbf{k})$ is the Hessian of a real-valued spring energy: real and symmetric for every $\mathbf{k}$. Real symmetric matrices admit real eigenvectors, and a real eigenvector bundle over any simply-connected BZ region is topologically trivial. Every plaquette holonomy reduces to the identity and every Berry/Wilczek–Zee curvature vanishes identically. We have verified this numerically across a full α sweep over the entire (k_x, k_y) plane: holonomy is identity to machine precision at every plaquette and every α.

The physical gauge connection instead lives over spacetime (x, t). It is the holonomy of the complex carrier envelope $\Psi \in \mathbb{C}^3$ transported around loops in real space and time, with the $U(1)$ trace read as the electromagnetic sector. The Brillouin zone and Bloch's theorem enter only to supply the carrier band and to justify the narrowband isolation of a wavepacket about its carrier $\mathbf{k}_0$; they are not the base space of the gauge connection.

The reason the base space is (x, t) and not a spatial slice is that the complex structure itself originates in the timelike direction: the $U(1)$ phase is the rotation of the real carrier field as one advances along the time axis (Section 5 of Paper I). A single spacelike slice is a real snapshot and carries no phase; two adjacent time slices are the minimum data that fix the $U(1)$.

3.6. Spinorial holonomy and the 4π aspect

For an $n = 2$ subspace (a two-level polarization bundle), holonomy lives in a double cover of $SO(3)$. A closed loop in physical space that induces a 2π rotation of an internal polarization frame can therefore produce a sign change of the transported state, while a 4π loop returns it to itself. This spinorial 4π aspect is attributed to Berry/Wilczek–Zee holonomy of the internal polarization bundle. Its connection to physical spin-$\frac{1}{2}$ phenomenology is part of the matter bridge (Paper IV).

3.7. Gauge bridge open problems

1. **Carrier-mode propagation law.** The gauge connection structure (A_μ, $F_{\mu\nu}$, covariant derivative) is established by the bridge. The outstanding task is to derive that free fluctuations of the Berry/Wilczek–Zee connection propagate as massless modes with Maxwell/Yang–Mills dispersion: the carrier-excitation spectrum in the appropriate limit should reproduce the massless spin-1 photon/gluon propagation law.
2. **Inhomogeneous Maxwell equations.** Show that the substrate furnishes the Maxwell action $-\frac{1}{4}\int F \wedge \star F$ (with its emergent Hodge star), so that the inhomogeneous equations $\mathrm{d}\star F = J$ hold with the correct coupling.
3. **Adiabaticity in the nonlinear regime.** Verify that the adiabatic condition (6) holds in the localized-mode parameter regime and quantify the non-adiabatic correction to the holonomy.

4. Color bridge: axis-triplet $U(3)$ structure and kinematic confinement

4.1. Axis-aligned triplet and prestress-driven degeneracy

On the 6-neighbor axial-only cubic lattice it is natural to organize the per-wavepacket complex envelope as a three-component triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$ aligned with the lattice axes. The prestress parameter α controls the eigenvalue spread of the 3-channel Christoffel matrix $D(k)$ without introducing off-diagonal coupling between axis components.

Non-degenerate limit ($\alpha \to 1$, no prestress). The lattice energy is purely Hookean longitudinal; $D(k)$ has maximally split eigenvalues. The three axis channels are dynamically decoupled, each supporting an independent rank-1 Berry connection $a_\mu^{(x)}, a_\mu^{(y)}, a_\mu^{(z)}$. The gauge group is closest to $U(1)^3$.

Degenerate limit ($\alpha \to 0$, maximum prestress). The geometric-stiffness term $(1-\alpha)|\Delta u|^2$ dominates and contributes equal stiffness to all components on every bond. $D(k) \propto I$ becomes proportional to the identity at every k; all three lateral channels are fully degenerate. The transported subspace is genuinely 3-dimensional and the adiabatic-transport object is a Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$ (Equation (7)). At the project's default operating point $\alpha = 0.2$ the lattice sits close to this degenerate limit.

Role of microscopic anisotropy. The $U(3)$ gauge group on a 3-dimensional complex envelope is a generic representation-theoretic fact about $\mathbb{C}^3$. What the cubic anisotropy provides is the *physically meaningful decomposition*: a preferred 3-dimensional subspace (the lateral triplet, distinct from X^4), a preferred axis-aligned basis within it (which makes the $SU(3)$ traceless generators carry operational meaning as "color" labels on solitons), and a natural identification of the $U(1)$ trace with the electromagnetism-like sector. Without cubic anisotropy, no preferred basis exists and the $SU(3)$ generators would be arbitrary linear combinations with no operational handle. That anisotropy can carry genuine non-Abelian gauge structure in real space is established by precedent: Chen et al. [2] demonstrate a synthetic $SU(2)$ gauge field for optical waves from the off-diagonal permittivity of an anisotropic medium.

4.2. $U(3) \simeq U(1) \times SU(3)$ *decomposition*

Decomposing $\mathcal{A}_\mu \in \mathfrak{u}(3)$ into trace and traceless parts:

$$\mathcal{A}_\mu = \underbrace{\tfrac{1}{3}\mathrm{tr}(\mathcal{A}_\mu)\, I_3}_{U(1)\text{ part}} + \underbrace{\sum_{a=1}^{8} A_\mu^a\, T^a}_{SU(3)\text{ part}}, \tag{12}$$

where T^a are the eight traceless $\mathfrak{su}(3)$ generators. The $U(1)$ part behaves electromagnetism-like; the traceless part carries non-Abelian self-couplings through the commutator term in $\mathcal{F}_{\mu\nu}$.

EM as the symmetric (trace) direction. The $U(1)$/trace component of Ψ is the projection onto $(1,1,1)/\sqrt{3}$:

$$\Psi_{\mathrm{EM}} = \frac{1}{\sqrt{3}}(\psi_x + \psi_y + \psi_z)\,\frac{(1,1,1)^\top}{\sqrt{3}}. \tag{13}$$

A pure-EM excitation displaces all three lateral components in phase with equal weight; the traceless complement ($SU(3)$) encodes differential phase and amplitude between components. A single-component displacement (e.g. $\psi_x \neq 0$, $\psi_y = \psi_z = 0$) projects as $\frac{1}{3}$ into the $U(1)$ trace and $\frac{2}{3}$ into the $SU(3)$-traceless sector; it is not a pure-EM excitation.

Charge-triality locking. Because $U(3) = \big(U(1) \times SU(3)\big)/\mathbb{Z}_3$ — the center $\mathbb{Z}_3 \subset SU(3)$ identified with the cube-roots of unity in $U(1)$ — single-valuedness forces the trace charge q and the $SU(3)$ triality $t \in \{0,1,2\}$ to satisfy $q \equiv -t \pmod 3$, the standard relation for $U(N) = \big(U(1) \times SU(N)\big)/\mathbb{Z}_N$ [3,4]. A color triplet therefore carries fractional ($\frac{1}{3}$-integer) trace charge while a color singlet carries integer charge — charge fractionality is locked to color representation without reference to dynamics.

Energetic sector coupling. The central-force anharmonicity (quartic coefficient $\propto k_s\alpha/a$, Equation (20) of Paper I) couples the two sectors at quartic order through a vertex $\propto \alpha|\Psi_{\mathrm{tr}}|^2|\Psi_\perp|^2$, which vanishes identically as $\alpha \to 0$. The prestress parameter that splits the two sectors' spectral scales is therefore also the parameter that couples them energetically.

4.3. *Topological structure of the two sectors*

The $U(1)$ and $SU(3)$ factors carry sharply different topology [5–7]:

- $\pi_1(U(1)) = \mathbb{Z}$ admits line vortex defects (codimension-2 in 3D), with phase winding by 2π around the core [8].
- $\pi_1(SU(3)) = 0$ forbids vortices in the $SU(3)$ sector.
- $\pi_3(SU(3)) = \mathbb{Z}$ admits Skyrmion/instanton-type textures (codimension-3 worldlines in 4D) [9,10].

The two sectors support topologically *different* classes of localized objects: codimension-2 vortex worldsheets in the EM/$U(1)$ sector versus codimension-3 texture worldlines in the color/$SU(3)$ sector.

However, because the full vacuum manifold is $U(3)$ — not just $U(1)$ — the $U(1)$ vortex is a *semilocal* vortex [11,12]: π_1 of the full $U(3)$ group is $\mathbb{Z}$ (the $SU(3)$ factor is simply connected), so the vortex is topologically protected within the $U(1)$ factor but not against perturbations that mix in the $SU(3)$ sector. Stability is therefore dynamical rather than purely topological, and may depend on the amplitude regime and the ratio of core energy to orientational moduli [13,14].

4.4. Kinematic confinement of color

A free linear triplet wavepacket propagating on the cubic lattice must satisfy two requirements:

Coherence. The three components stay phase-aligned over long range. This requires the three branch frequencies to be degenerate: $g(\hat{k}) = 0$, where $g(\hat{k}) = \sqrt{\sum_a (h_a - \bar{h})^2}/\bar{h}$ measures the spread of the lateral branch frequencies in direction $\hat{k}$. On the 6-neighbor cubic lattice, $g(\hat{k}) = 0$ holds *only* along the body diagonal $\hat{k} \parallel [111]$.

Color-activity. Operationally meaningful $SU(3)$-traceless holonomy — a finite fibre-internal rotation distinguishing the three axis components — requires the branches to be *non-degenerate*: $g(\hat{k}) \neq 0$, i.e. propagation *off* [111]. Exactly on [111] the degenerate triplet has no preferred internal basis and the color holonomy is undefined.

No linear propagation direction satisfies both simultaneously. Off [111] the unequal branch speeds dephase the triplet over long range; on [111] there is no resolvable color. Consequently:

> *Color charge cannot be carried to infinity by a free linear excitation. It has support only in a nonlinear, phase-locked bound state, where the three components are held together by the geometric coupling of Paper I rather than by spectral degeneracy.*

We call this *kinematic confinement of color*. It is a statement about asymptotic states (no colored free particle), not a dynamical derivation of a linear inter-quark potential, string tension, or area law.

Falsifiable handle. The off-[111] fibre-holonomy ratio scales with prestress as

$$\frac{R(\alpha_2)}{R(\alpha_1)} = \frac{(3-2\alpha_2)/\alpha_2}{(3-2\alpha_1)/\alpha_1}, \tag{14}$$

giving 0.3077 for $(\alpha_1, \alpha_2) = (0.2, 0.5)$. A deviation above $\sim 10\%$ would falsify the spectral-susceptibility factorization underlying it.

4.5. Numerical targets

The theory contains two distinct particle sectors with different topology and different numerical questions.

Target 1: $U(1)$ **carrier-phase vortex (electron-like sector).** The $U(1)$ trace sector supports line vortices. A preliminary rotating-frame-periodic JFNK relaxation of a pure-trace Y_1^1

vortex seed on a 112^3 lattice did not converge: the core drifted ≈ 19 lattice units off the symmetry axis, and the measured winding changed sign relative to the seed. This is the expected signature of semilocal orientational moduli with a dynamical $SU(3)$ sector. A converged eigenstate has not been obtained; branch-selection machinery for stabilizing the semilocal vortex remains future work.

Target 2: $\pi_3(SU(3))$ **texture (baryon-like sector).** The cubic substrate's axis-aligned triplet and $\pi_3(SU(3)) = \mathbb{Z}$ admit topologically stable Skyrmion-type textures. The canonical ansatz is the *hedgehog*

$$\xi^i(\mathbf{x}) \;=\; f(r)\,\hat{x}^i, \tag{15}$$

with $J = 0$, $L = 1$: internal index locked to spatial direction, invariant only under the combined diagonal $SO(3)$. The entire nontrivial structure lives in the $SU(3)$ traceless sector; the $U(1)$ trace averages to zero on the sphere, giving a trace-neutral far field (neutron-like). Topological stabilization ($B = 1$ winding) is achieved by extending the hedgehog into a Skyrme-twisted configuration. A nonzero $L = 0$ scalar admixture shifts the $U(1)$ far field to proton-like.

4.6. Color bridge open problems

1. **Verify holonomy ratio** (14) **numerically.** Compute the off-[111] fibre-holonomy at $\alpha = 0.2$ and $\alpha = 0.5$ and check the predicted ratio 0.3077.
2. **Converged** $U(1)$ **vortex eigenstate.** Develop branch-selection machinery to stabilize the semilocal vortex against orientational drift and obtain a converged periodic-BC eigenstate.
3. **Baryon/hedgehog stability test.** Construct the hedgehog ansatz on the cubic lattice, evolve under periodic-BC dynamics, and determine whether a stable, localized π_3-winding mode exists.
4. **Dynamical co-localization of** $U(1)$ **and** $SU(3)$ **sectors.** Determine whether the quartic coupling $\propto \alpha|\Psi_{\mathrm{tr}}|^2|\Psi_\perp|^2$ is sufficient to dynamically bind the charge to the color core in a baryon-like state.

Author Contributions: L.M. is the sole author and is responsible for all aspects of this work: conceptualization, formal analysis, software, writing—original draft, and writing—review and editing.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The substrate solver, diagnostic suite, and all experiments reported in this series are openly available at https://github.com/LukasMolzberger/BraneSim [15].

Acknowledgments: The author gratefully acknowledges Richard Behiel, Grant Sanderson (3Blue1Brown), and Dr. Jeroen Vleggaar (Huygens Optics) for educational content that inspired and informed this work.

Conflicts of Interest: The author declares no conflicts of interest.

https://doi.org/10.3390/sym1010000

Appendix A. Symbol dictionary

$\Omega \subset \mathbb{R}^3$	Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$).
$\mathbf{X}(x,t) \in \mathbb{R}^4$	Embedding field of the brane into ambient 4D Euclidean space.
t	Slicing parameter along the inside observer's timelike direction ($t := x^4$); not an external/ambient time (the ambient is symmetric 4D Euclidean, Section 3.1).
g_{ij}	Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$.
g^0_{ij}	Isotropic reference metric: $g^0_{ij} = \ell_0^2\, \delta_{ij}$ (stress-free scale ℓ_0).
$h_\star$	Held ground-state scale (e.g. imposed by boundary conditions).
α	Dimensionless pre-stress parameter: $\alpha = \ell_0/h_\star \in (0,1]$.
$\widehat{g}$	Mixed relative metric: $\widehat{g} = (g^0)^{-1} g$.
E	Green–Lagrange strain: $E = \frac{1}{2}(\widehat{g} - I)$.
λ, μ	Effective Lamé parameters derived from the spring constitutive law: $\mu(\alpha) = (1-\alpha)k_s/a$, $\lambda = 0$ on the 6-neighbor stencil. StVK agrees at quadratic order only; see Section 3.5.
ρ_m	Brane mass density in the Lagrangian.
$W(g; g^0)$	Isotropic stored-energy density (hyperelastic).
$P_A^{\ i}$	First Piola stress: $P_A^{\ i} = \partial W / \partial(\partial_i X^A)$.
$\mathbf{u}$	Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_\star$.
u	Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$.
$\boldsymbol{\epsilon}$	Normalized polarization state of a narrowband multi-component wave.
a_μ	Berry (rank-1) connection: $a_\mu = \mathrm{i}\langle u \,\vert\, \partial_\mu u \rangle$.
$f_{\mu\nu}$	Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$.
$\mathcal{A}_\mu$	Wilczek–Zee connection on an n-dimensional polarization subspace.
$\mathcal{F}_{\mu\nu}$	Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - \mathrm{i}[\mathcal{A}_\mu, \mathcal{A}_\nu]$.
γ_Γ	Geometric phase / holonomy around loop Γ: $\gamma_\Gamma = \oint_\Gamma a_\mu \, \mathrm{d}x^\mu$.
Ω_{mix}	Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.
$\Delta\omega_{\mathrm{gap}}$	Local spectral gap to the nearest competing polarization band.
a	Lattice spacing of the cubic substrate (microphysical length scale).
$\tilde{a}$	Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$).
$\mathbf{k}$	Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.
$\Psi = (\psi_x, \psi_y, \psi_z)^\top$	Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).
T^a	Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.
$R_p^l \in \mathbb{R}^4$	4D node position in the explicit lattice: $p = (i,j,k)$ spacelike triple, $l \in \mathbb{Z}$ timelike index.
k_s	Central-force spring stiffness of the 6-neighbor spacelike stencil.
m	Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$).
Δt	Temporal lattice step (timelike link length in the explicit 4D lattice).
$\mathcal{N}_s$	6-neighbor axial spacelike stencil: $\{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$.
$\mathcal{N}_t$	Temporal neighbor stencil: $\{l \pm 1\}$.
c_L	Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2/m$.
c_T	Long-wavelength transverse wave speed: $c_T^2 = (1-\alpha)\, k_s a^2/m$.

References

1. Xiao, D.; Chang, M.C.; Niu, Q. Berry Phase Effects on Electronic Properties. *Reviews of Modern Physics* **2010**, *82*, 1959–2007. https://doi.org/10.1103/RevModPhys.82.1959.
2. Chen, Y.; Zhang, R.Y.; Xiong, Z.; Hang, Z.H.; Li, J.; Shen, J.Q.; Chan, C.T. Non-Abelian gauge field optics. *Nature Communications* **2019**, *10*, 3125. Synthetic SU(2) gauge field from material anisotropy in homogeneous media; off-diagonal permittivity/permeability act as spin-dependent vector potentials; predicts Zitterbewegung trajectories and a non-Abelian Aharonov–Bohm effect with extractable Wilson loop, https://doi.org/10.1038/s41467-019-10974-8.
3. Georgi, H.; Glashow, S.L. Unity of All Elementary-Particle Forces. *Physical Review Letters* **1974**, *32*, 438–441. Georgi–Glashow $SU(5)$: the smallest simple group embedding $SU(3) \times SU(2) \times U(1)$; one unified coupling at M_G, https://doi.org/10.1103/PhysRevLett.32.438.
4. Baez, J.C.; Huerta, J. The Algebra of Grand Unified Theories. *Bulletin of the American Mathematical Society* **2010**, *47*, 483–552, [arXiv:hep-th/0904.1556]. Hypercharge normalization and the $\mathbb{Z}_6$ quotient $G_{SM}/\mathbb{Z}_6 = S(U(2) \times U(3)) \subset SU(5)$ underlying charge quantization, https://doi.org/10.1090/S0273-0979-10-01294-2.
5. Kibble, T.W.B. Topology of Cosmic Domains and Strings. *Journal of Physics A: Mathematical and General* **1976**, *9*, 1387–1398. Homotopy classification of defects: $\pi_0 \to$ walls, $\pi_1 \to$ strings, $\pi_2 \to$ monopoles, $\pi_3 \to$ textures, https://doi.org/10.1088/0305-4470/9/8/029.
6. Vilenkin, A.; Shellard, E.P.S. *Cosmic Strings and Other Topological Defects*; Cambridge University Press, 1994. Standard monograph on the $\pi_n(G/H)$ defect spectrum after symmetry breaking.
7. Manton, N.; Sutcliffe, P. *Topological Solitons*; Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2004. Standard reference for vortices, monopoles, skyrmions, instantons and the Atiyah–Manton construction, https://doi.org/10.1017/CBO9780511617034.
8. Nielsen, H.B.; Olesen, P. Vortex-Line Models for Dual Strings. *Nuclear Physics B* **1973**, *61*, 45–61. Abelian-Higgs vortex line; $\pi_1(U(1)) = \mathbb{Z}$ winding quantizes flux as $\Phi = (2\pi/e)n$ – the EM-like codim-2 line defect, https://doi.org/10.1016/0550-3213(73)90350-7.
9. Skyrme, T.H.R. A Unified Field Theory of Mesons and Baryons. *Nuclear Physics* **1962**, *31*, 556–569. Baryon as an $SU(2)$ π_3 texture; winding number identified with baryon number, https://doi.org/10.1016/0029-5582(62)90775-7.
10. Atiyah, M.F.; Manton, N.S. Skyrmions from Instantons. *Physics Letters B* **1989**, *222*, 438–442. Builds 3D π_3 skyrmion baryons from 4D $SU(N)$ Yang–Mills instanton holonomy, https://doi.org/10.1016/0370-2693(89)90340-7.
11. Vachaspati, T.; Achúcarro, A. Semilocal Cosmic Strings. *Physical Review D* **1991**, *44*, 3067–3071. Stable codim-2 vortex on a simply-connected (π_1 trivial) vacuum for $\beta < 1$: stability is dynamical, not topological – key precedent for an SU(3)-unfrozen electron vortex, https://doi.org/10.1103/PhysRevD.44.3067.
12. Achúcarro, A.; Vachaspati, T. Semilocal and Electroweak Strings. *Physics Reports* **2000**, *327*, 347–426, [arXiv:hep-ph/hep-ph/9904229]. Review; the $n = 1$ semilocal vortex carries an internal $CP^1 = SU(2)/U(1)$ orientational modulus – a $U(1)$ line defect inside a larger non-Abelian symmetry; β = type-I/II superconductor parameter, https://doi.org/10.1016/S0370-1573(99)00103-9.
13. Auzzi, R.; Bolognesi, S.; Evslin, J.; Konishi, K.; Yung, A. Nonabelian Superconductors: Vortices and Confinement in $N = 2$ SQCD. *Nuclear Physics B* **2003**, *673*, 187–216, [arXiv:hep-th/hep-th/0307287]. Non-Abelian vortex with $CP^{N-1} = SU(N)/[SU(N-1) \times U(1)]$ orientational moduli from color-flavor locking, https://doi.org/10.1016/j.nuclphysb.2003.09.029.
14. Eto, M.; Hamada, Y.; Nitta, M.; et al. Dynamics of Global and Local Vortices with Orientational Moduli. *Journal of High Energy Physics* **2021**, *2021*, 156, [arXiv:hep-th/2012.04098]. $U(1)$-gauged + global $SO(N)$ vortex with core-localized S^{N-1} orientational Nambu–Goldstone moduli, https://doi.org/10.1007/JHEP03(2021)156.
15. Molzberger, L. BraneSim: a deterministic brane-lattice substrate simulator. https://github.com/LukasMolzberger/BraneSim, 2026. Open-source substrate solver, diagnostic suite, and experiment pipeline reproducing the figures in this paper.